

Supplementary Material: When Vision Goes Dark

Anonymous CVPR submission

Paper ID *

1. Backbone Generality: ViT-B/14

The main paper reports the late-layer CKA collapse on ViT-S/14 and cites the ViT-B/14 replication. Here are the full numbers. Two observations go beyond the summary in the main text. First, ViT-B/14 is more robust at the cliff — 36.7% at severity 4 against ViT-S’s 23.3% — yet lands at the identical floor (10.8% at severity 5): backbone capacity buys mid-ladder robustness, not floor escape. Second, ViT-B’s grace regime is flat rather than bumped (98.3% at both severities 0 and 1), so the bump is not a universal property of the family but depends on capacity or scale — a distinction the full-scale run can settle.

Table 1 gives the ViT-B/14 response curve; Table 2 the per-block CKA. The late-minus-early drop gradient steepens from +0.32 (ViT-S) to +0.44 (ViT-B), and the largest single drop moves to a later block, consistent with the depth-compounding account.

Table 1. ViT-B/14 main curve (pilot scale): accuracy and embedding drift.

Sev.	ViT-B/14 acc (%)	cos to clean
0	98.3	1.00
1	98.3	0.95
2	92.5	0.80
3	68.3	0.52
4	36.7	0.24
5	10.8	0.13

2. Seed and Regularization Sensitivity

The main curve regenerates across three split seeds ($\{42, 43, 44\}$) crossed with three logistic-probe regularization strengths ($C \in \{0.1, 1, 10\}$) — nine full re-runs of the ladder per severity. Table 3 reports mean $\pm$ standard deviation. The regime structure (never-negative grace, cliff, chance-level floor) is present in every configuration; the cliff’s exact position varies by at most

Table 2. ViT-B/14 layer-wise CKA to clean (test fold).

Block	CKA sev3	CKA sev5	Drop (0→5)
0	0.932	0.823	0.177
1	0.864	0.626	0.374
2	0.801	0.599	0.401
3	0.633	0.466	0.534
4	0.538	0.324	0.676
5	0.578	0.333	0.667
6	0.577	0.262	0.738
7	0.596	0.263	0.737
8	0.604	0.258	0.742
9	0.638	0.210	0.790
10	0.592	0.143	0.857
11	0.543	0.155	0.845

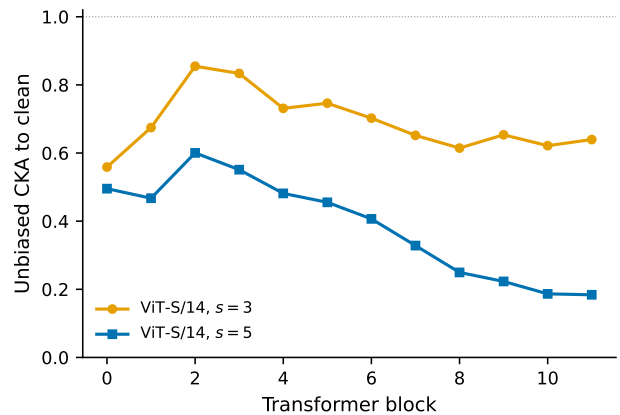


Figure 1. CKA to clean by block: ViT-S/14 (solid) vs. ViT-B/14 (dashed, severity 5). The drift gradient deepens with backbone size.

one severity step.

Table 3. Accuracy (%) across 3 split seeds $\times$ 3 probe-C values.

Severity	Accuracy (mean $\pm$ std, $n=9$)
0	92.2 $\pm$ 1.3
1	93.6 $\pm$ 1.1
2	89.4 $\pm$ 3.1
3	67.3 $\pm$ 7.1
4	27.2 $\pm$ 5.0
5	10.9 $\pm$ 1.5

3. Failure Analysis Detail

Per-class death order. Table 4 gives per-class accuracy at severities 0/1/3/5. Automobile, cat, dog, and truck cross the 50% line at severity 3; airplane, bird, deer, horse, and ship hold until severity 4. Frog never crosses it — the constant-class artifact analyzed in the main text (§6.4 there). Note also that the classes dying early (automobile, truck) are exactly those whose clean embeddings sit closest to their dark-counterpart embeddings in cosine terms, suggesting embedding proximity, not semantic difficulty, governs the death order; we flag this as an observation for the full-scale run to test, not a claim.

Table 4. Per-class accuracy (%) across the ladder.

Class	Clean	Sev 1	Sev 3	Sev 5 (%)
airplane	92	92	85	8
automobile	100	100	33	0
bird	100	89	56	0
cat	83	92	42	0
deer	91	100	73	0
dog	85	85	38	0
frog	92	100	92	100
horse	93	93	80	0
ship	100	100	62	0
truck	100	100	46	0

Persistence. Every image misclassified at severity 1 remains misclassified at severity 5 (100/100 on the pilot fold). A hundred clean-correct images flip by severity 5.

4. Efficiency

Latency measured on CPU (batch 1) with interleaved rounds to cancel machine-contention drift: 4 rounds $\times$ 25 iterations per arm, median per arm. Adapters add 21–25% latency for at most 1.18M trainable parameters.



Figure 2. Clean-correct images misclassified at severity 5 (title shows the dark prediction). Most dark predictions are “frog” — the readout collapse of the main text.

Table 5. Latency and throughput by arm.

Arm	Backbone (M)	Trainable (M)	Latency (ms)	Acc (dark, %)
frozen_backbone	22.1	0.00	40.6	56.2
lora_r4	22.4	0.29	49.0	–
lora_r8	22.6	0.59	50.9	–
lora_r16	23.2	1.18	50.7	–

5. Corollary Tests: Prior vs. Geometry, Readout vs. Enhancement

The main text reports that the constant-class collapse survives a five-fold oversampling of a non-frog class (geometry attractor, not prior replay) and that readout-side repair beats the best input-space enhancement. This section gives the full statistics (Table 6).

Table 6. Floor statistics of the fixed probe at severity 5 under three training-prior conditions (identical test fold and backbone). The collapse target does not follow the prior.

Training prior	Top-1 class	Share	$H/\ln 10$
near-uniform (default)	frog	0.99	0.02
class-balanced weights	frog	0.99	0.02
cat oversampled $\times 5$	frog	0.99	0.02

Readout repair vs. the best input-space enhancement (linear gain), paired permutation test on the same fold: +15.8 points at $s=4$ ($p=0.007$), +20.0 points at $s=5$ ($p<10^{-3}$).

6. Prediction-Distribution Statistics

Per-severity statistics of the fixed probe’s output distribution (Table 7): top-1 predicted class and its share, number of classes ever predicted, and Shannon entropy of the prediction distribution normalized by $\ln 10$ (uniform = 1.0). The collapse to a constant class at severity 5 (entropy 0.02, two classes used) is the mechanism behind frog’s spuriously perfect per-class accuracy.

Table 7. Prediction-distribution statistics of the fixed probe by severity.

<i>s</i>	top-1 class	share	classes used	$H/\ln 10$
0	horse	0.12	10	0.99
1	horse	0.12	10	0.99
2	ship	0.12	10	0.99
3	frog	0.28	10	0.91
4	frog	0.72	10	0.50
5	frog	0.99	2	0.02